

Restaurant Management System:

Project Description:

The restaurant management system is a console-based application in C++. It manages restaurant operations like customers and menu items. Users can add, update, delete and view menu items and customers orders. This system uses file handling for persistence data storage.

Project objects:

- Manage customer information efficiently.
- Maintain menu records.
- Allow adding, updating, deleting and viewing menu items.
- Demonstrate practical C++ concepts such as OOP, file handling and vectors.

System Modules:

1. Menu Module:

Manage the restaurant menu. Allows adding, Updating, deleting and view menu items.

Menu Data Structure:

```
struct Menu  
  
{  
  
    int id;  
  
    int price;  
  
    string name;  
  
    double prize;  
  
};
```

Operations:

- Add Menu Item.
- Delete Menu Items.
- Update Menu Item.
- View Menu Items

File storage: menu.csv

2. Customer Module:

Handles customer operations such as adding new customers, viewing details, updating and deleting customer records.

Customer Data Structure:

```
struct Customer  
  
{  
  
    int id;  
  
    string name;  
  
    string phone;  
  
    string address;  
  
};
```

Operations:

- Add Customer
- Delete Customer
- Update Customer
- View customer details.

File storage: Customer .csv

Order Module:

Tracks customer orders and maintains a transaction history.

Order Data structure:

```
struct Order  
  
{  
  
    Int id;  
  
    String name;
```

```
int quantity;  
  
}
```

Operations:

- Place order
- View order history.
- Generate Bill

File storage: order.csv

Bill Module:

Calculate total amount for a customer's order and prints a detail bill with items, quantities and prices.

```
struct Bill  
  
{  
  
    Int item;  
  
    double price;  
  
}
```

Operations:

1. Select customer
2. Add ordered items with quantities.
3. Calculate subtotal for each item.
4. Calculate total bill amount.
5. Print formatted bill.

File handling: bill.csv

Technologies Used

- Programming Language: C++
- File Handling for data storage.
- Console-based user interface using cout and cin

- Data structures: struct and vector

Conclusion:

This restaurant management system is a practical way to learn C++ concepts such as:

- Object -oriented programming with struct.
- File Handling for persistent data storage.
- Console-based user interface using cout and cin
- Vector and dynamic data management.